

Supplementary Information

Agentic diagnosis of lower respiratory tract infection from metagenomic sequencing and multimodal clinical data

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Supplementary Tables

Table S1: **mNGS organism features computed before scoring.** Source: `upstream/tools/rank_mngs_candidate_microbes.py`, `upstream/tools/mngs_common.py`, `deterministic_mngs_max_scorer.py` (`dominance_by_record`).

Feature	Values	Definition
Control-subtracted priority	1–5	Rank of each reported organism in the laboratory’s negative-control comparison (1, no signal in any negative control; 5, weakest), as output with the mNGS report. Code score 1.0, 0.85, 0.65, 0.45, 0.30.
Read percentile	0–1	Percentile of $\log_{10}(\text{reads} + 1)$ among all organisms in the run.
Final score / label	high, medium, low	$0.6 \times \text{code score} + 0.4 \times \text{percentile}$; high ≥ 0.70 , medium ≥ 0.45 .
Read tier	R0–R4	R0: 1–9 reads; R1: ≥ 10 ; R2: ≥ 100 ; R3: $\geq 1,000$; R4: $\geq 10,000$.
Dominance tier	D0–D3	Per specimen record; top organism: D3 if reads $\geq 10 \times$ runner-up, D2 if $\geq 3 \times$, else D1; every other organism D0.
Specimen alignment	aligned, sterile/systemic, non-aligned	From the mNGS-to-specimen assignment and the target site (lower respiratory tract).
Hospital support	per module	Same-organism support from culture, FilmArray/galactomannan, molecular microbiology; “Support”, “Not support” or “Unknown”.
Host context	V0–V3, O0–O2	Host-vulnerability tier and opportunistic-coverage level; host expansion when V2–V3 or O1–O2.

Table S2: **Signal tiers and causative levels of the deterministic scorer (v20).** Source: `deterministic_mngs_max_scorer.py`, functions `m1_strong`, `m2_moderate`, `m3_protected`, `m4_weak`, `set_candidate_level_from_tier`, `level3_pickable`, `pickable`. Conditions are evaluated in order; the first satisfied tier applies.

Tier / level	Condition
M1 strong	priority 1–2 and R3–R4; or priority 1–2 and percentile ≥ 0.85 ; or D3 and $\geq R2$ and priority ≤ 3 .
M2 moderate	priority 1–2 and R2 and percentile ≥ 0.75 ; or D2–D3 and $\geq R2$ and priority ≤ 3 ; or same-organism auxiliary support and $\geq R1$ and priority ≤ 3 ; or protected pathogen (not reactivation virus or mould), $\geq R1$, priority ≤ 3 , label not low; or R3–R4 and priority ≤ 4 and label not low; or <i>P. jirovecii</i> with R1–R2 and priority ≤ 3 ; or Mucorales at priority 1 with percentile ≥ 0.60 . Reactivation viruses and <i>Aspergillus</i> use dedicated clauses.
M3 protected	retention condition of a protected pathogen; or priority 1–2 with R0–R2; or priority ≤ 3 with percentile ≥ 0.60 above R0; or priority ≤ 3 with high/medium label and R1–R2; or priority 3 with R1–R2 and non-host auxiliary support; or priority 4 with R4.
M4 weak	priority 4–5; or label low; or R0–R1; or non-aligned specimen.
M5 background	organism on the background list.
Level 1	M1, aligned or sterile specimen, and (D3 or R3–R4).
Level 2	other M1; M2 with aligned or sterile specimen.
Level 3	M2 non-aligned; M3; M4 under host expansion when plausible (not reactivation virus, colonizer or non-aligned; with non-host support, protected status, or priority ≤ 3 with R1 and high/medium label).
Level 4	other M4; any organism demoted by a guardrail.
Level 5	M5.
Picked	level 1–2; level 3 with protected status, an oral-flora/respiratory-commensal representative rule, related typical-pneumonia context, direct support from culture, FilmArray/galactomannan or molecular testing, or (host expansion) priority ≤ 2 with $\geq R2$ or D2–D3; <i>Candida</i> /generic yeast never at level 3 without invasive support. Fallback: level-4 “best available” when nothing is pickable.

Table S3: **Guardrail rule identifiers of the deterministic scorer.** All 33 R-S4-* identifiers in `deterministic_mngs_max_scorer.py`. Descriptions summarize the identifier and rule body.

Identifier (R-S4-...)	Effect
BACKGROUND	background organism, level 5
CANDIDA-BLOOD-STERILE-CONTEXT	<i>Candida</i> retained on blood/sterile-site evidence
CANDIDA-HOSPITAL-NONINVASIVE-NOT-PICKED	<i>Candida</i> with only non-invasive hospital support not picked
CANDIDA-RESP-CONTEXT-NOT-PICKED	respiratory <i>Candida</i> context not picked
CANDIDA-RESP-L3	respiratory <i>Candida</i> held at level 3; picked only with invasive support
COMMON-HOSPITAL-R0R1-NONDOMINANT-PICKED-OMIT	common hospital organism with R0–R1, non-dominant: omitted from picks
GENERIC-UNIDENTIFIED-HOSPITAL-LABEL	generic labels (e.g. “Gram-negative bacilli”) never picked
HERPES-HOST-R2R3-NONDOMINANT-REVIEW	herpesvirus with host-only support, R2–R3, non-dominant: review only
HERPES-R4-NONDOMINANT-HOST-ONLY-REVIEW	herpesvirus with R4 but non-dominant and host-only: review only
IMPOSSIBLE-INFECTION-SOURCE	organism cannot be a pulmonary infection source
L3-MNGS-ONLY-HOST-LOW-NONDOMINANT	level-3 mNGS-only organism with low, non-dominant reads not picked
LOW-ACTIONABILITY-VIRUS-PICKED-OMIT	EBV, HHV-6/7 and similar omitted from picks
LOW-PULM-SPECIFICITY-ATYPICAL-L4	atypical organism of low pulmonary specificity, level 4
LOW-PULM-SPECIFICITY-ENV-L4	environmental organism of low pulmonary specificity, level 4
LOW-SPECIFICITY-VIRUS-L4	low-specificity virus, level 4
MNGS-ONLY-NONDOMINANT-NON-TYPICAL-PICKED-OMIT	mNGS-only, non-dominant, non-typical pneumonia organism omitted
MNGS-R0R1-HOST-ONLY-NONPROTECTED-PICKED-OMIT	R0–R1 with host-only support, non-protected: omitted
NON-CORE-RESP-VIRUS-PICKED-OMIT	non-core respiratory virus omitted from picks
NONPULMONARY-CULTURE-CONTEXT-ONLY	culture at a non-pulmonary site is context only
ORAL-ASPIRATION-FLORA-L3	oral/aspiration flora representative retained at level 3
ORAL-ASPIRATION-FLORA-MNGS-ONLY-NONDOMINANT-L4	mNGS-only, non-dominant oral flora, level 4
RARE-YEAST-HOSPITAL-MNGS-L3	rare opportunistic yeast with hospital and mNGS support, level 3
RARE-YEAST-MNGS-ONLY-CONTEXT	rare yeast with mNGS only: context
RELATED-REPRESENTATIVE-HOSPITAL-CONTEXT	related-organism hospital context supports a representative
RESP-COMMENSAL-L3	respiratory commensal retained at level 3
RESP-COMMENSAL-L4	respiratory commensal, level 4
SAME-REPRESENTATIVE-GROUP-PICKED-OMIT	one representative per taxonomic group; others omitted
SKIN-ENV-YEAST-L4	skin/environmental yeast, level 4
SKIN-LOW-SPECIFICITY-BACTERIA-L4	skin flora of low specificity, level 4
TRACE-PROTECTED-WATCHLIST-PREFERRED	trace-read protected pathogen kept on a watch-list, not picked
WATER-ENV-GNB-L4	water-borne environmental Gram-negative, level 4
WATER-ENV-LOW-SPECIFICITY-PICKED-OMIT	water-borne organism of low specificity omitted from picks
WEAK-MOLD-HOST-ONLY-PICKED-OMIT	weak mould signal with host-only support omitted

Table S4: **Evidence-module thresholds used by R5.** Sources:
RAG_re/config/default.json, RAG_re/config/formal_merge_full_a.json,
RAG_re_casefit/config/default.json, RAG_re_clinical/config/clinical_v1.json,
OBER_patient_delivery/generate_r5_final_decisions.py.

Module	Setting
A retrieval	PubMed E-utilities; query "organism"[Title/Abstract] AND (site terms in title/abstract); publication cutoff 2026-06-30; at most 10 articles; 0.34 s between requests; literature also retrieved for upstream picks (<code>literature_on_baseline_picked = true</code>).
A judgement	support requires exact species, human clinical evidence, target-site match and a valid evidence span (≤ 20 contiguous words); positive when ≥ 3 valid supports among ≥ 5 judgeable articles; otherwise abstain.
A1 case-fit	per-article verdict strong / partial / mismatch / insufficient (Methods); aggregation thresholds in config: ≥ 2 strong for high, ≥ 1 strong or ≥ 3 partial for moderate, ≥ 5 judgeable articles. R5 uses strong ≥ 2 or strong+partial ≥ 7 .
B	absolute axis: M1/M2 or R3/R4; relative axis: dominance $\geq D2$, or priority ≤ 1 with percentile ≥ 0.8 ; site alignment required; the optional negative-control-ratio and reads-per-million gates are disabled (<code>require_ntc_ratio = false</code> , <code>require_rpm = false</code>).
C	positive modules: culture, FilmArray, FilmArray/galactomannan, targeted molecular, molecular microbiology, PCR, antigen, galactomannan; grades C3 / C2 / C1 / C0 / CNEG as in Methods; respiratory <i>Candida</i> with culture only $\rightarrow$ C1.
D	pass / guarded / block / unknown; reported, not gated.
Temporal	window 3 days before and after the anchor (research arms only).
R5	picked OR C3 OR ((A1_strong ≥ 2 OR A1_match ≥ 7) AND site_aligned AND (B_absolute OR B_relative)); rule id R5_A1_UNION_AND_B_LOOSE.

Table S5: **Language models, prompts and configurations by stage.** SHA-256 prefixes of the files in the code repository at the time of writing (2026-09-20).

Stage	Model / setting	File	SHA-256 (prefix)
Record standardization (raw sections → standardized JSON)	GPT-5.6 Luna	upstream/core/llm_parser.py	–
Slide-image transcription (image input)	GPT-5.6 Luna, Responses API, image detail high	reference_versions/shared_upstream/ppt_patient_parser/vision_hooks.py	–
Host/laboratory agent	GPT-5	agents/prompts/CBC_Lab_underlying_prompt.txt	0929360cfb5d
Culture agent	GPT-5	agents/prompts/culture_agent_prompt.txt	e1631735b119
FilmArray and GM agent	GPT-5	agents/prompts/filmarray_GMtest_prompt.txt	2a092516ff13
Imaging agent	GPT-5	agents/prompts/image_agent_prompt.txt	b0ea51281d8f
Molecular agent	GPT-5	agents/prompts/molecular_microbiology_prompt.txt	802ccf52ac71
mNGS-to-specimen	GPT-5 or deterministic v4	agents/prompts/mNGS_to_specimen_prompt.txt	643dcf157308
Deterministic summary	rules v3 (reported) / v4 (current)	tools/build_deterministic_summary.py; build_deterministic_summary_v3_recovered.py	–
Scorer	rules v19 (reported) / v20 (current)	tools/deterministic_mngs_max_scorer.py	–
Safety review of unpicked organisms	GPT-5.6 Luna, reasoning medium	prompt in tools/review_missed_mngs_candidates.py	–
Post-review rules	deterministic: name canonicalization, weak-environmental omission, non-pulmonary-support guardrail, syndrome-pattern and tier convergence, atypical-serology rescue, group representatives	tools/merge_deterministic_max_with_missed_review.py	–
Article judge (A)	GPT-5.6 Luna, reasoning low, 1,200 tokens	RAG_re/prompts/article_support_v1.txt; config/default.json; formal_merge_full_a.json	af4178902c50; bc123c20c743; b3ca56742dce
Case-fit (A1)	GPT-5.6 Luna, reasoning low, 1,400 tokens	RAG_re_casefit/prompts/article_casefit_v1.txt; evidence_span_repair_v1.txt; config/default.json	e495aef60640; f6f15feca9d9; a9f387850b01
Clinical rules (B, C, D)	clinical_v1	RAG_re_clinical/config/clinical_v1.json	41121351788c
R5	rule R5_A1_UNION_AND_B_LOOSE	OBER_patient_delivery/generate_r5_final_decisions.py	–
Rationale	GPT-5.6 Luna, reasoning low	generate_llm_rationales.py	–
Direct prompting	gpt-5.6-sol/luna/terra, gpt-6-astra; reasoning medium; 4,000 tokens	LLM_test/direct_raw_benchmark/prompt_v1.txt	961087b8acdf

Supplementary Notes

Supplementary Note 1. Missed-candidate safety-review prompt

The review prompt is embedded in `upstream/tools/review_missed_mngs_candidates.py` (`build_prompt`). The model receives the review queue, deterministic output, hospital summary and mNGS specimen assignments and returns one of four categories: `review_high_priority`, `review_context_needed`, `review_low_specificity` or `omitted_with_reason`. It may not replace deterministic selections or upgrade an organism because of immunocompromise or protected status alone. Respiratory *Candida* and generic yeasts are considered low-specificity without same-species invasive evidence, and different *Candida* species are not treated as aliases. CMV, HSV and VZV are classified as reactivation without evidence of primary pulmonary disease; EBV is not considered a pulmonary candidate. Additional rules address non-dominant herpesvirus signals, oral and aspiration flora, water-borne Gram-negative bacteria, gastrointestinal pathogens, coagulase-negative staphylococci and corynebacteria. The output is a JSON object with Traditional-Chinese rationale fields.

Supplementary Note 2. Direct-prompting baseline prompt

File `LLM_test/direct_raw_benchmark/prompt_v1.txt` (SHA-256 prefix 961087b8acdf), original in Traditional Chinese. English rendering:

You are an infectious-diseases clinical reviewer. Using only the single patient's data provided here, decide which microorganisms are most likely the clinical causative pathogens of the current pulmonary or lower-respiratory-tract infection episode.

This is the Direct-Raw baseline: do not use external search, PubMed, RAG, AETIA, any candidate pool or any earlier model result. The raw mNGS list may contain contamination, colonization, background organisms, low-level detections and organisms unrelated to the site of infection; "detected" does not mean "pathogen". Judge from the mNGS reads and the laboratory's control ranking, the specimen site, cultures, molecular tests, imaging, host status and timing. Respiratory *Candida*, low-specificity environmental organisms, skin flora and latent or reactivated viruses must not be listed as the infection source merely because they were detected. A positive result from a non-pulmonary specimen must not be treated as a pulmonary source without a pulmonary link. An empty array means "no record available", not a negative test. You may select zero, one or several pathogens; do not force an answer. Put only the organisms you finally judge to be the infection source in `predicted_pathogens`; if evidence is insufficient, put them in `excluded_detected_organisms` or `limitations`. Name pathogens at species level; do not merge different *Candida* or other species. Output must follow the given JSON schema; supporting evidence, counter-evidence, reason and limitations should be short and auditable, and hidden reasoning must not be output.

Supplementary Note 3. Case examples

The cases in Fig. 4e,f originated from the 2026-01-17 presentation of the first pipeline version in a KMH intensive-care series. In case 1, *Klebsiella pneumoniae* was dominant in the bronchoalveolar specimen (68,569 reads) and was selected in the presence of new bilateral consolidation, pleural effusions, increasing C-reactive protein and worsening hypoxaemia. *Candida albicans*

from the same specimen was classified as airway colonization because of low, non-dominant reads, repeated low-burden respiratory culture and the respiratory-*Candida* guardrail. In case 2, *Candida tropicalis* met the C3 direct-evidence criterion because several blood-culture sets grew the same species and the bronchoalveolar read burden was high.

The full record delivered for each organism contains: the disposition (selected or not) and the rule path that produced it; the value of modules A, A1, B, C and D; the per-modality findings recorded as support or opposition; the PubMed identifiers and judged spans of the case-matched articles; the specimen, site and timing used; and the SHA-256 hashes of the input files. [Replace these prototype vignettes with the delivered R5 records (selected organisms, rule paths, module results, audited explanation, reference labels) and obtain approval for case-level presentation.]

Supplementary Note 4. Version boundaries and replay

The 55 reported R5 decisions used rule version v19 (`mngs_deterministic_level_only_v19_low_specificity_taxonomy`). The recovered v19 scorer reproduced all saved scorer outputs from the stored summaries and ranked mNGS records. The current scorer, v20 (`mngs_deterministic_level_only_v20_central_taxonomy_profile`), uses a central rule-based organism classifier, removes a retired language-model scoring branch and retains M1 for typical pneumonia bacteria with lower-respiratory culture or FilmArray support where v19 assigned M2. Replaying the same inputs reproduced all candidate names, selected sets, merge-stage selections and review-tier assignments. Candidate levels changed in 15 patients and selection order changed in five. The clinical-summary replay used current patient directories rather than a verified frozen agent snapshot; reconstructed hospital evidence differed from the stored summaries for three patients, without changing any reported selection. Sources: `docs/UPSTREAM_REPLAY_SUMMARY.json`, `docs/UPSTREAM_RECOVERY.md` and `docs/VERSIONS.md`.

Supplementary Note 5. Evaluation outputs to be added

[Per-patient tables (predicted and reference sets per method), per-hospital metrics, the 9 rescued organisms with correctness, error analysis of false negatives and false positives, confidence intervals.]

Supplementary Note 6. Registry analysis of the sequencing report alone

The shared registry contained adjudicated pathogen lists for 33 cases (Methods; `tables/make_discordance.py`; `figures-source/discordance.csv`). mNGS reported 104 organisms, of which 38 (37%) were adjudicated as causative and 66 (63%) as colonizing, contaminating or incidental. The read-count distributions overlapped across five orders of magnitude. Read count ranked a causative organism above a non-causative organism in 61% of pairs (AUC 0.61), with AUCs of 0.73 for bronchoalveolar lavage fluid and sputum and 0.56 for blood, cerebrospinal fluid and tissue. The largest count, 189,467 reads of *Acinetobacter nosocomialis* in lavage fluid, was adjudicated as colonization despite culture growth. *Pneumocystis jirovecii* with 7 reads in another lavage specimen was adjudicated as causative on the basis of a positive lavage PCR.

Ten species were causative in one patient and non-causative in another: cytomegalovirus, Epstein–Barr virus, herpes simplex virus 1, *P. jirovecii*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *A. nosocomialis*, *Burkholderia cenocepacia*, *Candida albicans* and *Candida tropicalis*. The list includes airway colonizers and recognized pathogens from multiple organism classes. Species identity and a uniform read threshold did not replace patient-specific clinical context, particularly for non-sterile respiratory specimens.

Evidence from other modalities improved discrimination. An exact-organism conventional test corroborated 47% of causative detections and 18% of non-causative detections. Seventeen of 55 adjudicated pathogens (31%) were absent from the mNGS report of the sequenced specimen; 16 (94%) were identified by another test, including culture in eight, PCR in five, serum galactomannan in two and IgM serology in one ([figures-source/missed_by_mngs.csv](#)). Fourteen of these tests were collected within two days of the mNGS specimen, and the longest interval was five days. Eleven pathogens were identified in a different specimen type, and five were identified in the same specimen type by a different testing principle. These findings indicate complementary failure modes of sequencing and conventional tests. [This analysis covers the registry cases whose adjudication is recorded in the shared workbook; the KMH line keeps its adjudication in the per-patient workbooks and is not included. Recompute over all 105 specimens once those labels are consolidated.]